



Cavalry Bank

a mobile banking app

Yuyang Ding

Project overview



The product:

Cavalry Bank is a fictional mobile banking app based primarily on a leading UK bank. Users can manage their bank accounts, view card information, and make transfers within the app.



Project duration:

Jan.2026-Apr.2026



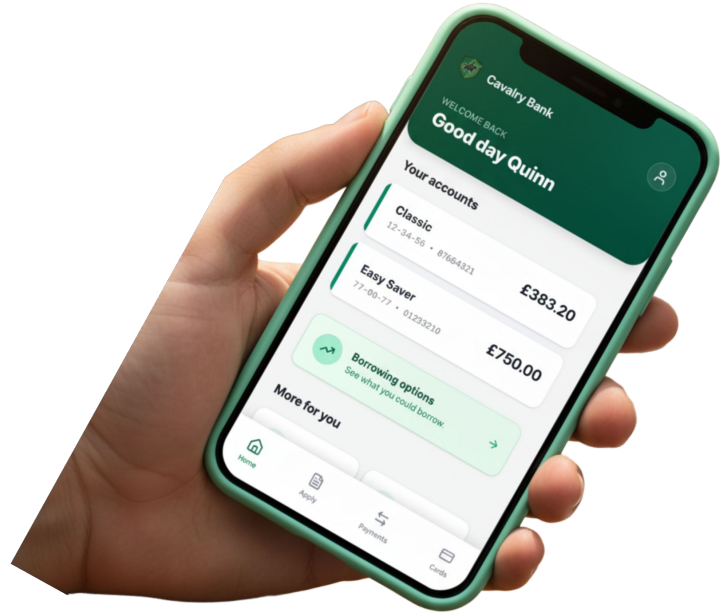
My role:

UX Researcher



Responsibilities:

User research; Prototyping; Usability Testing



Project overview



Background:

UK banks are changing.

Instead of visiting a local branch, many users prefer managing their finances on their phone.

High street banks launched their mobile banking app and kept improving the user experience in the past decade. While managing personal finance on a smartphone sounds cool, not everyone is very comfortable with it.....



Project overview



The problem:

Most functions are hidden deeply under accordion menus and/or require multiple steps to get into the right place. While users don't use most of the functions very often, it's very difficult for users to find a new function. Some pages have unclear instructions. Results of clicking on certain buttons tend to be unpredictable.



The goal:

Make sure the users can use the product to manage their money safely, confidently, and seamlessly.

This should at least include: opening, viewing, and closing accounts; viewing card details; making bank transfers.

Understanding the user

- Personas
- User journey maps
- User flow paradigm

User research: summary



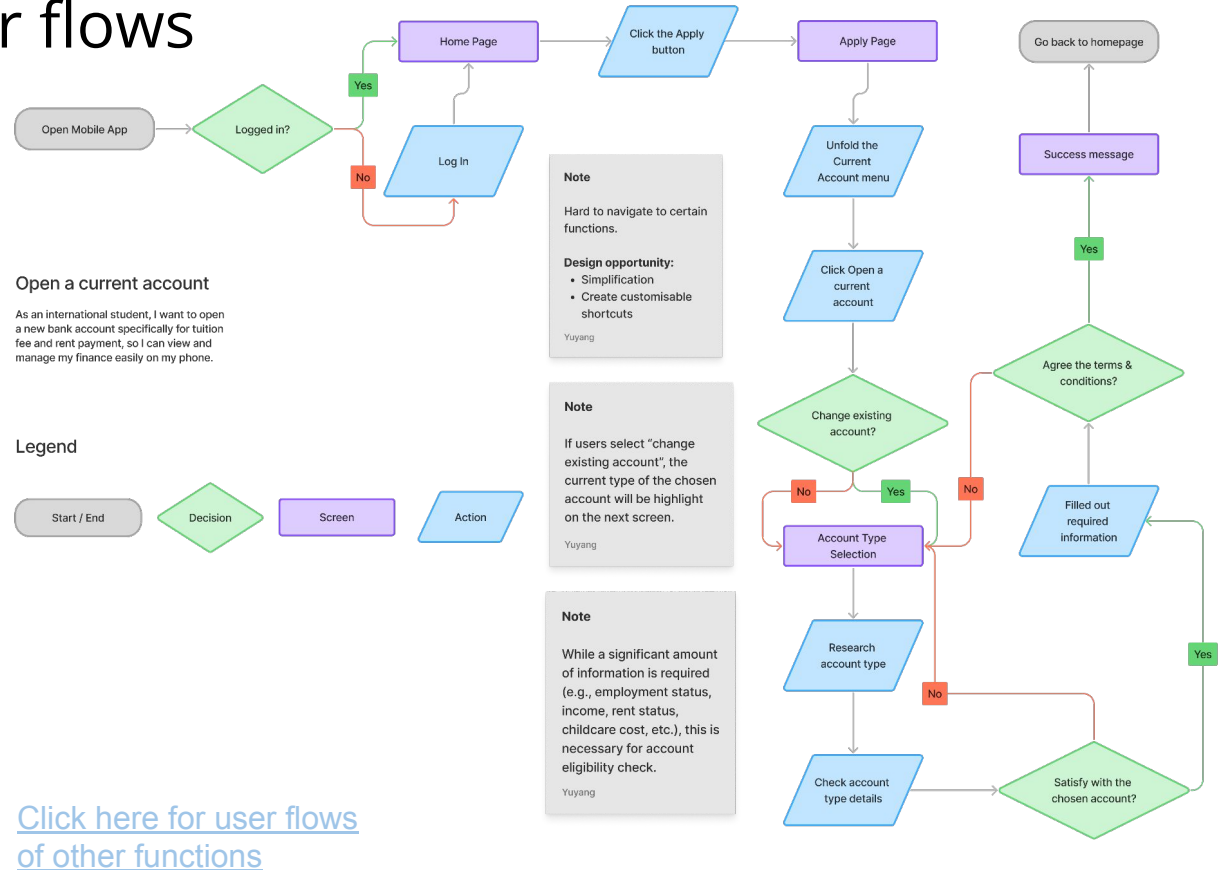
I began with emphasising my users: “How would I use this app? How would I feel about the user flow?”. By writing down the user flow, several potential pain points popped up.

Focusing on users with particular characteristics, a usability test was conducted with 5 participants to test whether the users can manage their individual finance smoothly with Cavalry Bank and understand users’ pain points.

User research: user flows

Take account opening for example. It's surprising (and annoying) that users need 3 clicks just to get to the first step of opening a new current account.

Even worse, this is also true for most functions. It's somewhat understandable, as most of the functions are not used very often, but are there better ways to do this?



[Click here for user flows of other functions](#)

User research: persona



After going through the user flows, it also came to my attention that several functions including account opening include a huge amount of questions to answer, forms to fill out, etc., through which many jargons and terms appears.

Most of the time, users with limited literacy or digital skills may struggle using products like mobile apps. International students and new immigrants, however, despite (usually) having both decent literacy and digital skills, can also struggle using a mobile banking app as they are not familiar with the local finance system, especially the jargon.

So, here comes May, an international student just arrived in the UK a couple of weeks ago...



May Zou

Age: 23

Education: Master's degree

Hometown: Tianjin, China

Family: Single

Occupation: Full-time student

“The banking system seems different here, I need to figure out how to manage my money safely.”

Goals

- Apply for a new current account to manage her money without visiting local branch.
- Make bank transfers and manage accounts smoothly and flawlessly.

Frustrations

- Not familiar with terms and jargon widely used in the British banking industry.
- Anxious about taking the wrong move - “What if I didn’t pay my tuition fee to the correct account?”

May is a 23-year-old international student from China. She has recently moved to the UK to pursue her master’s degree. Her English is academic-fluent, and she has good digital skills. However, she’s not very familiar with the banking system here, nor those jargons (what is a “standing-order”?).

Mei needs a UK bank account for both daily expenses and large expenditures like paying tuition fees and rent. She’s already used to doing all this stuff using mobile payment apps (e.g., PayPal, Alipay) instead of using banks. She would be comfortable if she were aware of the consequences of any actions on the mobile banking app.



May:

“I need to open a UK account to make my tuition fee payment securely.”

Step 1. Determination

Actions:

Review daily financial behaviors;
Assess if a UK bank account is necessary or make life easier.

Emotions:

“I have a credit card issued by my bank in China, but not sure if I can use it for all my expenses. There are also too many different banks: What is a high street bank? Shall I go with one of them?”

Improvement:

A mobile banking app with essential functionalities clearly presented, so that users can easily figure out if their needs can be met. Available accounts/cards options should also be advertised as clearly as possible.

Step 2. Discovery & Selection

Actions:

Download the Cavalry Bank app;
Register a user account;
Research account options.

Emotions:

“I’m lost and confused. It wasn’t easy to find the ‘open a new account’ button. God...there’s too many different types of accounts, which one do I need?”

Improvement:

Avoid hiding critical functionalities in deep menus. Put customisable shortcuts on the homepage;
Provide a short, jargon-free quiz (e.g., “What brings you to the UK?”) that recommends the most suitable account.

Step 3. Application

Actions:

Select prospected account;
Fill out the questions for validating the eligibility.

Emotions:

“There are TOO many questions; some of them are confusing, but I filled them out anyway. I also need to upload a proof of my student status...Hmmm, can I get one from my university?”

Improvement:

Only ask necessary questions and make sure the questions are related to the account the users applied to;
Provide detailed guidelines which are easily accessible right next to the relevant blanks.

Step 4. Agreement & Next

Actions:

Agree the Terms & Conditions;
Sign the required documents.

Emotions:

“Yah! Seems like I’m almost there. Wait, what happened? I clicked on “Agree”, and now it’s finished. Where is the final confirmation?”

Improvement:

A final confirmation with all relevant details presented clearly is needed, so that users can understand they are finishing.

Testing the old design

- Low-fidelity prototype
- Usability test

Testing the old design: summary



Thanks to the variety of financial services provided by the banks, modern mobile banking apps tend to feature tons of functions. However, a mobile banking app, essentially, should be something people use to manage their personal finances, as they have been doing at their local branch for many years.

Thus, I tried to replicate the key user interface of the this mobile banking app in Figma, and focused the testing on a short list of key functionalities of a mobile banking app: account management, payment, and card management.

Testing the old design: lo-fi prototype

[Click here for the full lo-fi prototype](#)



CAVALRY BANK

User Name / Email:

Password:

[Forgotten Your Password?](#)

Login

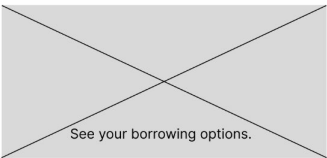
Sign Up

Good day Quinn

Your accounts

Classic
12-34-56/87654321£383.20

Easy Saver
77-00-77/01233210£750.00



See your borrowing options.

More for you



Plan my future



Everyday offers



Credit score



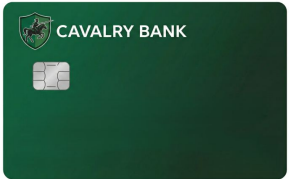
Explore travel hub

Home

Apply

Payments

Cards



Classic

£0 per month

Benefits of the account

✓ Free everyday banking

When you stay in credit, other fees may apply e.g. using your Visa debit card outside the UK.

✓ Discover our mobile banking app

With: spending insights, request a payment, card controls, credit score and more.

✓ Option for an arranged overdraft

Overdrafts are subject to our assessment of your circumstances and are repayable on demand.

✓ Contactless Visa debit card

✓ Pay on your phone

Use Apple or Google Pay with your debit card to make payments.

Select

Pay & Transfer

From:

Classic
12-34-56/87654321£383.20

To:

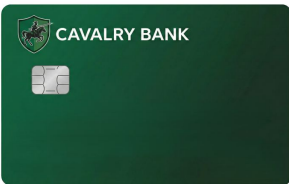
Cat

90-45-90/22223333

Amount:

Continue

Card management





Card Details



View PIN

Home

Apply

Payments

Cards

Testing the old design: usability test



The participants:

5 international students or prospective international students who haven't lived in UK before or less than 1 year.



Test duration:

Approximately 30 minutes per session



Key Performance Indices:

1. Time on tasks
2. User error/drop-off rates
3. System Usability Scale



Methods:

Moderated usability test (remote), participants take the test on their own phone while being accompanied with the UX researcher.



Tasks:

1. Login to the app and open a classic current account.
2. View the details (i.e., account number, sort code, PIN) of the card associated with the new account.
3. Initiate a bank transfer to a new payee.

Testing the old design: usability test

	1	2	3
	Open an account	View card details	Bank transfer
Time	average ~8.6 min	average ~1.5 min	average ~4.5 min
Error	5 out 5 fail to find the entrance on their first go	0 out 5 fail to find the entrance on their first go	1 out 5 got lost in the account detail page
SUS	difficulty 8 out 10	difficulty 2.4 out 10	safety 6.8 out 10

Testing the old design: usability test

1

Pain point

Functionalities hidden under complex menus. Users need to click multiple times and search amongst tens of categories to navigate to the entrance.

2

Pain point

Too many accounts types are provided in a single screen.

3

Pain point

Users feel unsure about the consequence of some decisions. On the very last screen of account opening procedure, the “Agree” button plays a role of submission and complete the account opening immediately without giving users the chance to do a final review.

4

Pain point

No authentication verification before critical actions like viewing card PIN, confirming transfer, etc.

Refining the design

- Paper wireframes
- High-fidelity prototype
- Usability studies

Paper wireframes

Let's start from grabbing a pen.

Collecting everything we already know: persona, users' pain points, and their user flows.

Now, the question is how can we improve their experience?

Before making changes, I derived several potential solutions with pen & paper.

The wireframes are organized into four columns, each representing a different screen in a banking application. The first column, titled 'Apply', shows a form with fields for 'Function Category' and 'Entrprises', and a section for 'replacing accoridions with icons?'. The second column, titled 'Classic Account', shows a form for 'Yuyang VISA' with a 'Benefits' section and an 'Apply' button. The third column, titled 'Account application', shows a form for 'The legal bit' with a 'T&C' section and a 'Submit' button. The fourth column, titled 'Account application', shows a form for 'Final confirmation' with a 'Confirm' button. Arrows indicate the flow between screens, such as from 'Apply' to 'Classic Account' and from 'Account application' to 'Final confirmation'. A large arrow points from the first column to the second, and another from the third to the fourth. A third arrow points from the first column to the fourth. A fourth arrow points from the first column to the third. A fifth arrow points from the first column to the fourth. A sixth arrow points from the first column to the fourth. A seventh arrow points from the first column to the fourth. An eighth arrow points from the first column to the fourth. A ninth arrow points from the first column to the fourth. A tenth arrow points from the first column to the fourth. A eleventh arrow points from the first column to the fourth. A twelfth arrow points from the first column to the fourth. A thirteenth arrow points from the first column to the fourth. A fourteenth arrow points from the first column to the fourth. A fifteenth arrow points from the first column to the fourth. A sixteenth arrow points from the first column to the fourth. A seventeenth arrow points from the first column to the fourth. An eighteenth arrow points from the first column to the fourth. A nineteenth arrow points from the first column to the fourth. A twentieth arrow points from the first column to the fourth. 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Apply

XXXX: Function Category

Entrprises

XXXX:

XXXX:

XXXX: Open accounts

replacing accoridions with icons?

Classic Account

XXXXXX

Yuyang VISA

Benefits:

xx xx

xx x

xx xx

Apply

(everyone applys for basic accounts and allow for upgrading later)

Account application

The legal bit:

T&C

XXXXXX

XXXXXX

XXXXXX

① By tickiny the box below

XXXXX

XXXXX

XXXXX

☒ I have read and agree the T&Cs above

Submit

Account application

Final confirmation:

XXXXXX

XXXXXX

XXXXXX

XXXXXX

Classic account

Once you press the "confirm" button below we will send you a new card.

Confirm

Good day Yuyang!

(Home)

Your accounts:

XX-XX-XX

£100,000

XX-XX-XX

£5,000

Open Accounts

Shortcuts

customis your shortcuts

Account advisor

Why do you need a new account?

① XX-XX-XX

② XX-XX-XX

③ XX-XX-XX

④ XX-XX-XX

Next

(recommend one based on users' answers to a few questions)

Pay & Transfer

Quick Payment

Auto-addl quick payees

① ② ③

Aaron Bella Cat

From:

XX-XX-XX

£100,000

To:

① Choose who to pay

Confirm Payment

Does this look right?

Enter your PIN

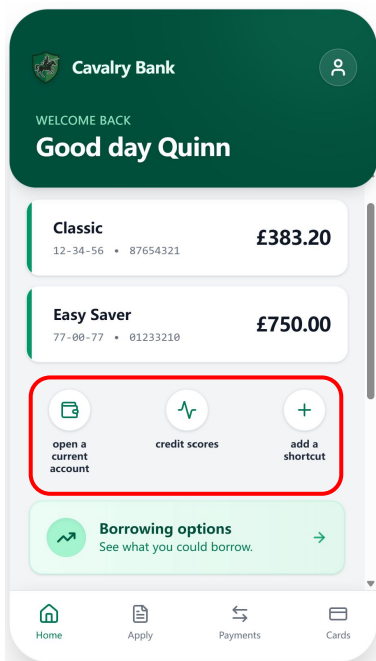
1	2	3
4	5	6
7	8	9
0	⌫	⌂

add a final verification

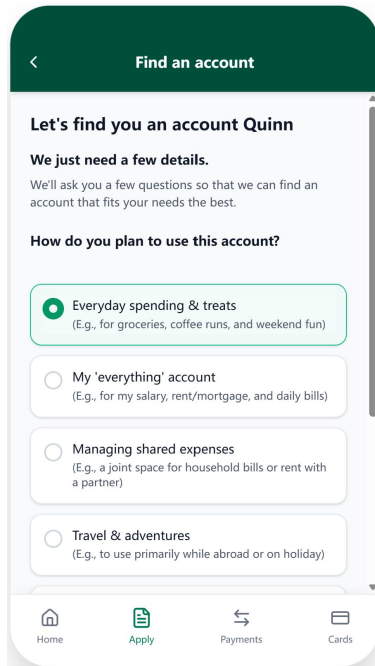
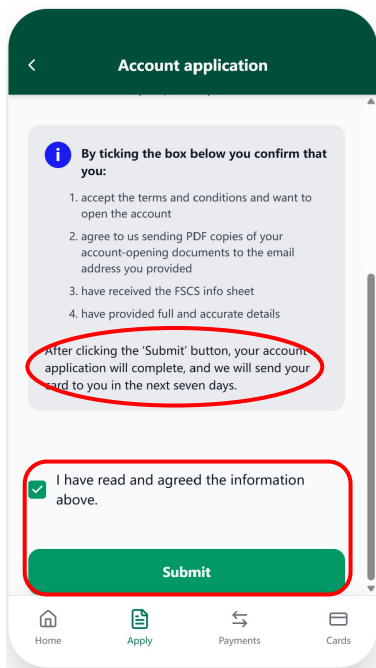
face ID

Refining the design: hi-fi prototype

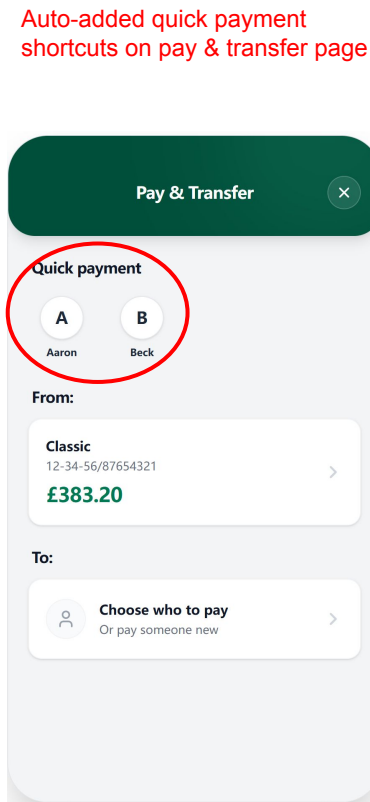
[Click here for the full hi-fi prototype](#)



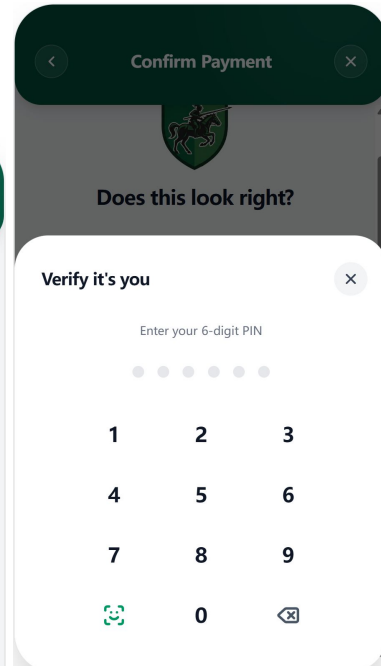
Replace “Agree” with “Submit” + checkbox and highlight the consequence of clicking the button



Add an account advisor, which help users find the account they need through a few questions



Auto-added quick payment shortcuts on pay & transfer page



Add a final verification with PIN or faceID before confirming a transfer

Add customisable shortcuts on the homepage

Refining the design: usability test

	1	2	3
	Open an account	View card details	Bank transfer
Time	average ~4.6 min	average ~1.5 min	average ~4.1 min
Error	0 out 5 fail to find the entrance, although still complain about too many forms to fill	0 out 5 fail to find the entrance on their first go	0 out 5 got fail to make the payment properly
SUS	difficulty 6.1 out 10	difficulty 2.2 out 10	safety 8.0 out 10

Going forward

- Takeaways
- Next steps

Takeaways

Mobile banking is special:

For a mobile bank, it's almost inevitable that a significant amount of functions are zipped into a single App, and users' behaviour and preferences can vary largely (e.g., Judy may even have no idea about the existence of a function frequently checked by Jack). Thus, it's really hard to figure out which function should be highlighted. For apps/websites like a mobile bank, which features too many functions, it is better to allow users to create shortcuts to their frequently-used functions.

Safety is king, but how safe is safe?

There is no doubt that safety is the top requirement of a banking app, but users may have different levels of safety requirements. For a UK user, it's probably normal to enter the password/use FaceID only once at login. This can be scary for a user who lived elsewhere for a long time and resides in the UK recently: "What do you mean by I can pay ALL my money to another guy without entering my PIN?" Not all apps are about safety, but it is important to bear in mind that users with different background can have different feelings and needs. It's important to empathise all potential users.

Experience-Business trade off:

It is somewhat understandable that banks want to use every chance to advertise their products (e.g., premier accounts), but users can get overwhelmed by too many options. Presenting 5 different accounts to users in one go may not be the best idea, at least not provide users with the best experience. Instead of filling users' screen with various products, it might be better to use a brief questionnaire or even utilising Large Language Models to help users find the accounts (or any other products) they need.

Next Steps

While I'm confident that the refined design could provide smoother experience, there are still unsolved problems.

Filling forms is exhausting

Even with a shortcut on the home page, users still find the account-opening difficulty, mainly due to the long procedure. This is a tougher one. Yes, the procedure is very long (10 steps). No, we can't just get rid of them. Some steps are about the legal bit, so have to keep; some others are about verifying the users' eligibility for certain types of accounts. In the next iteration, it might worth put some attention on the latter: is it possible to simplify the procedure for, at least, applicants to a basic-level account?

Beyond account-opening

The issue with opening an account is quite common and also apply to many other functions. Even worse, many users may only open one or two account with a bank, but they can use other functions more often. It is important to iterate over other functions and try the best to provide smooth and safe user experience.

AI + UX?

This is kind of a side product of this UX Research project.

As a PhD candidate in an experimental discipline, I am already very used to using LLMs in my research: from debugging my code for online experiments and analysing data (yes, psychology is also a heavy-quantitative discipline), to grammar checking of my manuscript. And now, I find another field where LLMs can do an impressive job: prototyping.

Of course, you can't use them like a Wish Machine, and for a better outcome, you still need to create a lo-fi prototype with all the necessary details yourself, or at least a sketch. Then, send it to Figma (or other vibe coding tools, but Figma allows you to copy your prototype into the chat directly) with clear prompts, after a few minutes, TA-DA! You got an (almost) usable hi-fi prototype!

If you have some basic knowledge of Typescript or feel comfortable coding with the assistance of LLMs, then this is a fairly good starting point. With some minimal work, you can easily build a decent hi-fi prototype on top of this, as I presented in this project. How amazing!

Definitely give it a try, if you haven't before. Use it wisely, then LLMs will become one of your most powerful tools!